FIAP

Mobile App Development

View, Flexbox e Style

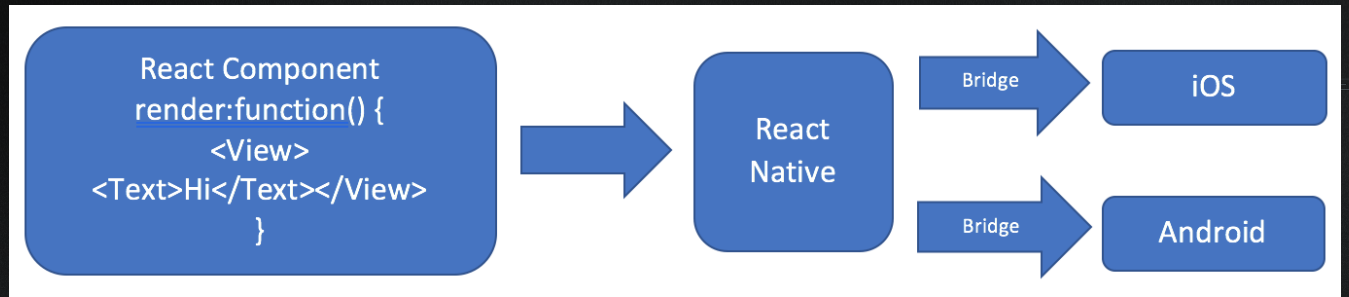
SLIDER

View

View

O que é a View?

É o componente básico do React-native. É um container já declarado com layout Flexbox e que na compilação é traduzido para o container da respectiva tecnologia.



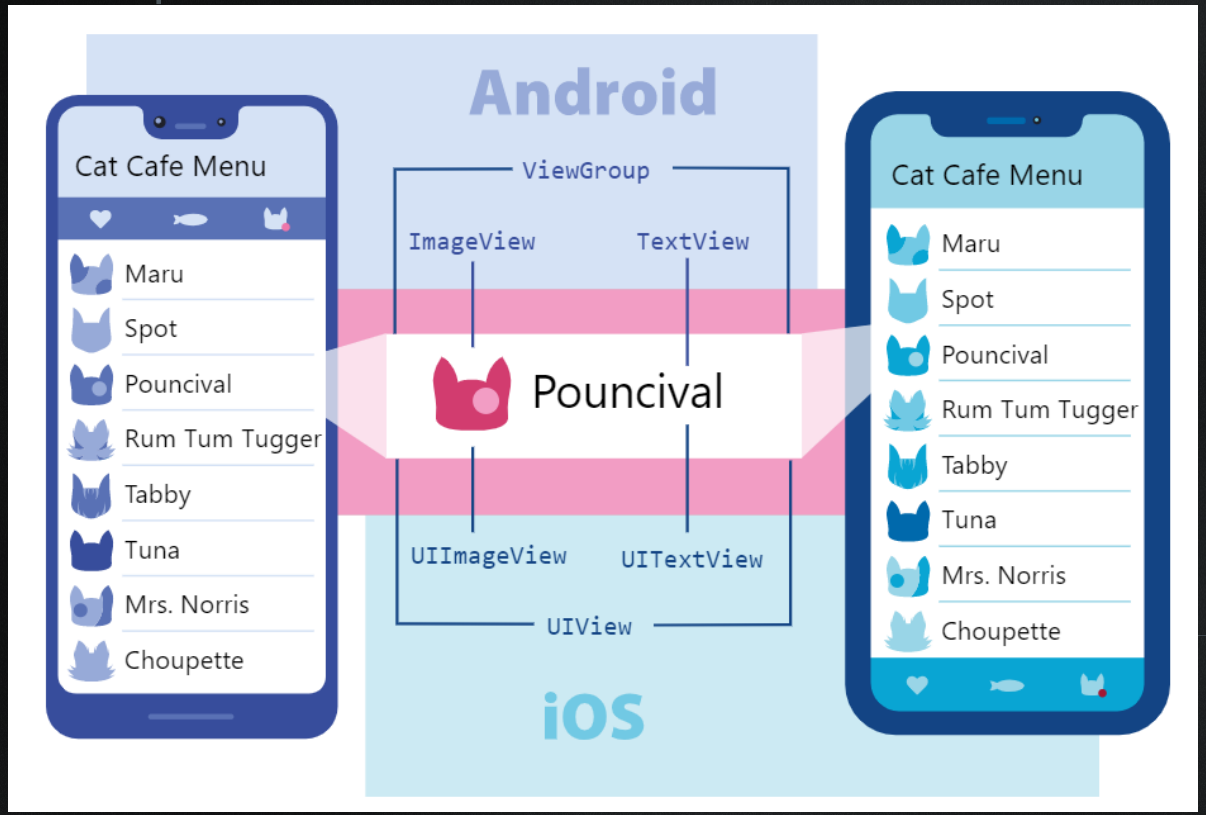
View

View Function Component Example

Expo

```
import React from 'react';
import {View, Text} from 'react-native';

const ViewBoxesWithColorAndText = () => {
  return (
    <View
      style={{
        flexDirection: 'row',
        height: 100,
        padding: 20,
      }}>
      <View style={{backgroundColor: 'blue', flex: 0.3}} />
      <View style={{backgroundColor: 'red', flex: 0.5}} />
      <Text>Hello World!</Text>
    </View>
  );
};

export default ViewBoxesWithColorAndText;
```

View


Android
Cat Cafe Menu
Maru
Spot
Pouncival
Rum Tum Tugger
Tabby
Tuna
Mrs. Norris
Choupette
ViewGroup
ImageView
TextView
Pouncival
UIImageView
UITextView
UIView
iOS
Cat Cafe Menu
Maru
Spot
Pouncival
Rum Tum Tugger
Tabby
Tuna
Mrs. Norris
Choupette

FlexBox

Flexbox

O que é o Flexbox?

É um modelo de **layout** criado para conseguirmos organizar componentes de maneira **dinâmica**, simples.

É muito utilizado pela sua adaptabilidade da disposição e tamanho dos componentes, tornando uma ótima escolha para designs responsivos

Flexbox

Entendendo o conceito básico

A organização do flexbox se dá pela divisão dos componentes entre containers e itens.

É muito utilizado pela sua adaptabilidade da disposição e tamanho dos componentes, tornando uma ótima escolha para designs responsivos

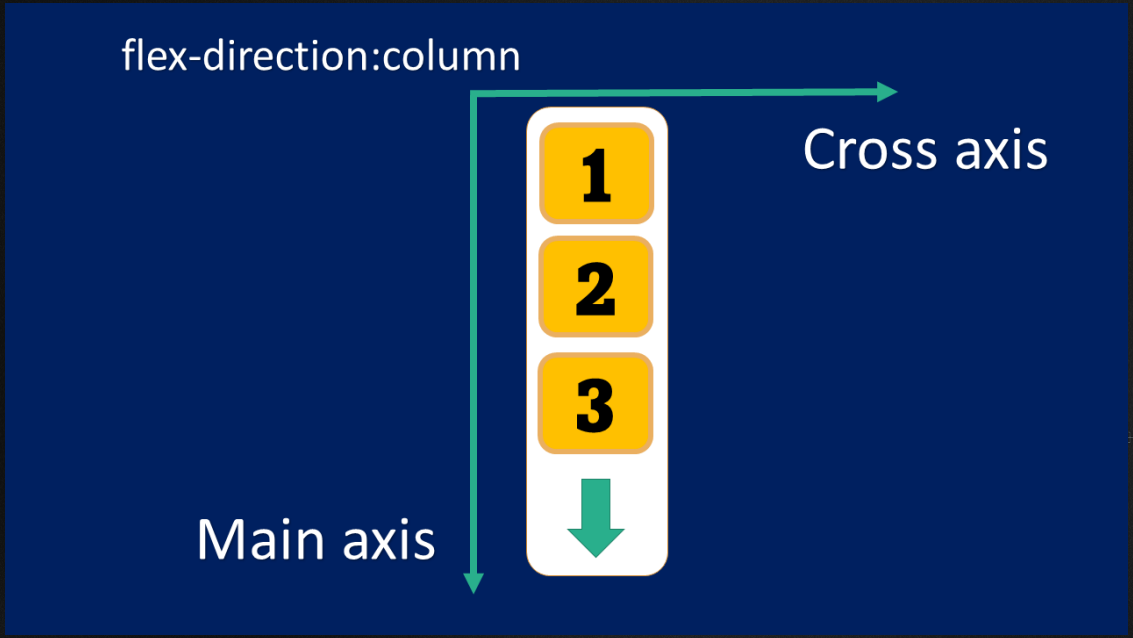
Flexbox

Nomenclaturas importantes

- Eixo principal
- Eixo transversal
- Item
- Container

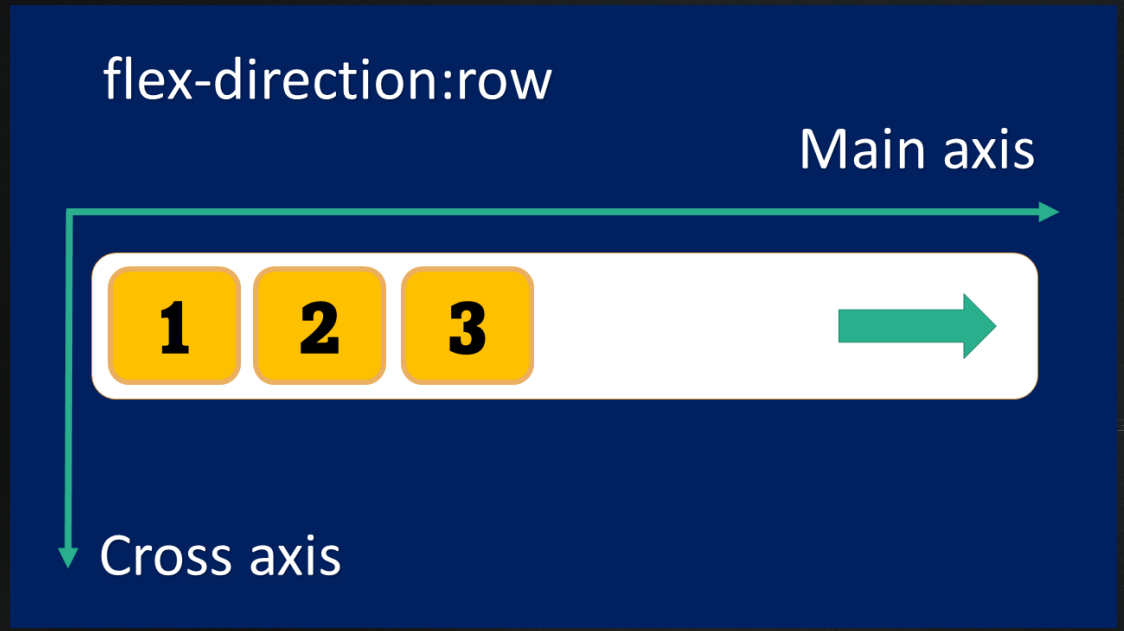
Flexbox

Nomenclaturas importantes



Flexbox

Nomenclaturas importantes

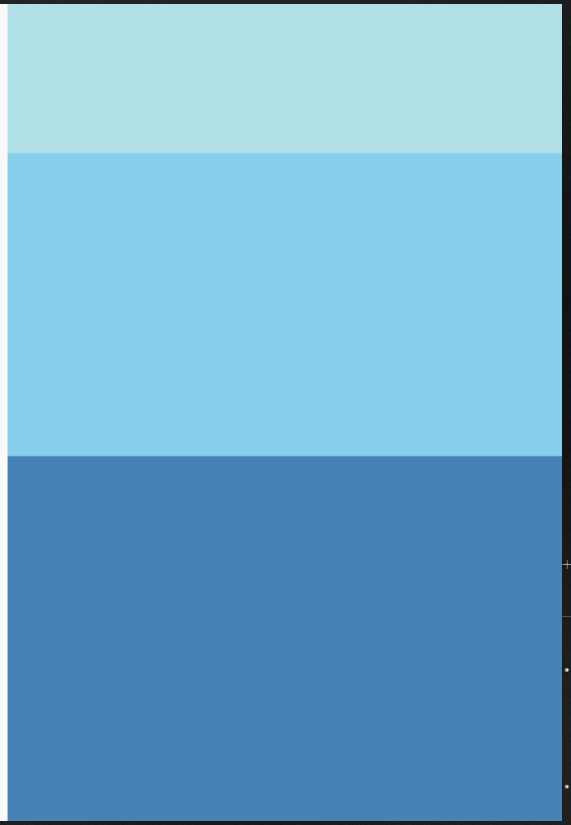


View

```react-native
import React from 'react';
import {View, Text} from 'react-native';

const ViewBoxesWithColorAndText = () => {
  return (
    <View
      style={{
        flexDirection: 'row',
        height: 100,
        padding: 20,
      }}>
      <View style={{backgroundColor: 'blue', flex: 0.3}} />
      <View style={{backgroundColor: 'red', flex: 0.5}} />
      <Text>Hello World!</Text>
    </View>
  );
};

export default ViewBoxesWithColorAndText;
```

View

```react-native
import React from 'react';
import {View} from 'react-native';

const FlexDimensionsBasics = () => {
  return (
    // Try removing the `flex: 1` on the parent View.
    // The parent will not have dimensions, so the children can't expand.
    // What if you add `height: 300` instead of `flex: 1`?
    <View style={{flex: 1}}>
      <View style={{flex: 1, backgroundColor: 'powderblue'}} />
      <View style={{flex: 2, backgroundColor: 'skyblue'}} />
      <View style={{flex: 3, backgroundColor: 'steelblue'}} />
    </View>
  );
};

export default FlexDimensionsBasics;
```



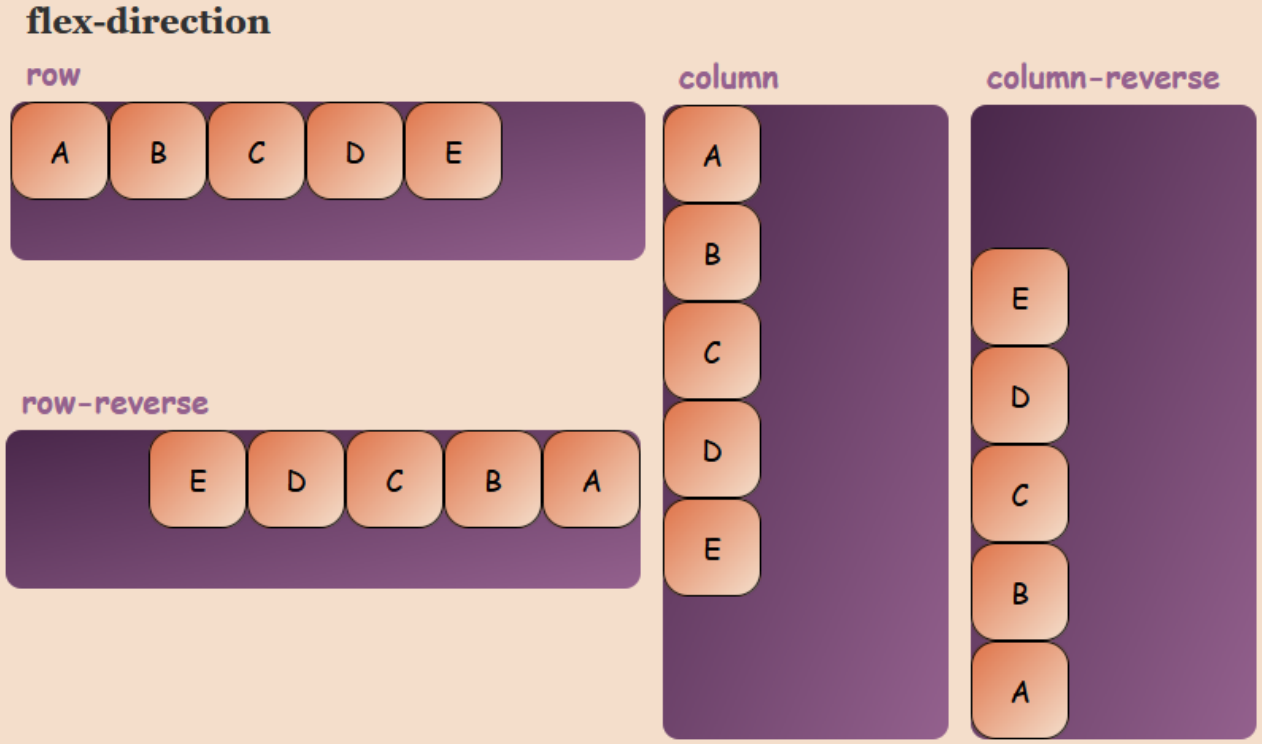
Flexbox

flexDirection

Determina a direção e sentido que os itens são inseridos

- row
- row-reverse
- **column (default do componente View)**
- column-reverse

Flexbox


flex-direction
row
A
B
C
D
E
column
A
B
C
D
E
column-reverse
E
D
C
B
A
row-reverse
E
D
C
B
A

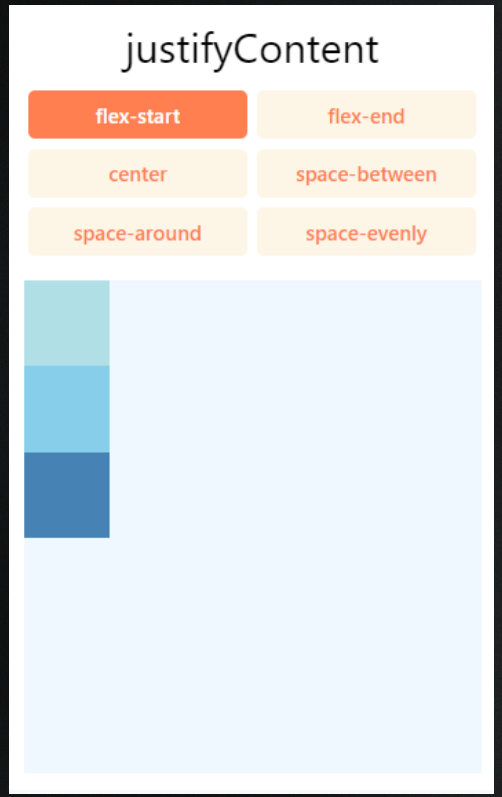
Flexbox

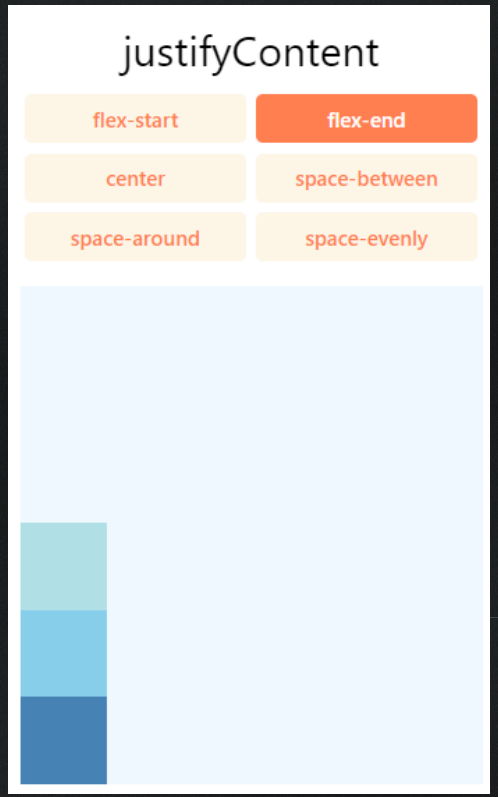
justifyContent

Determina a distribuição dos itens dentro do componente.

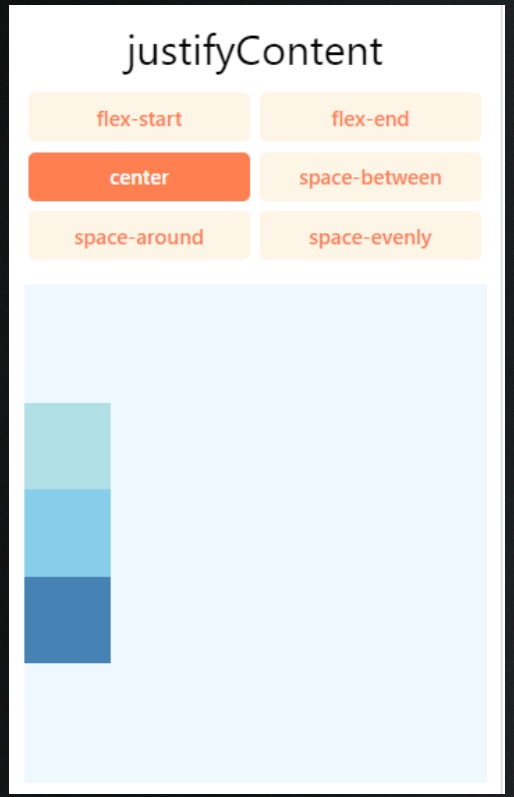
- **flex-start (default)**
- flex-end
- center
- space-between
- space-around
- space-evenly

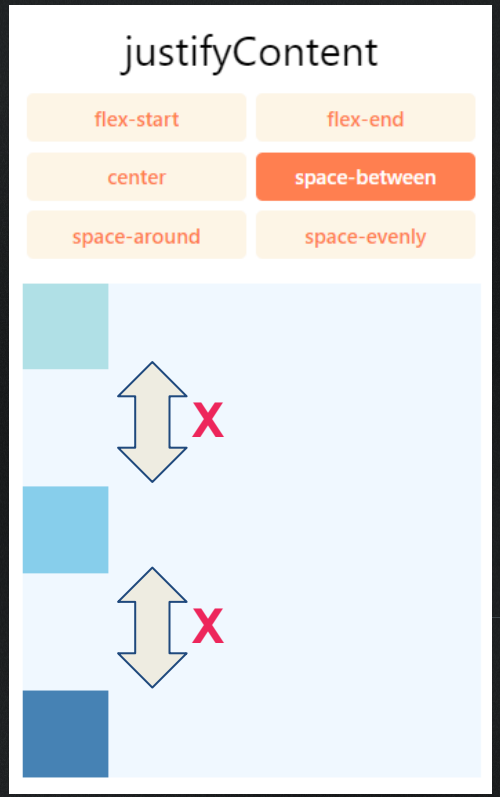
Flexbox


justifyContent
flex-start
flex-end
center
space-between
space-around
space-evenly

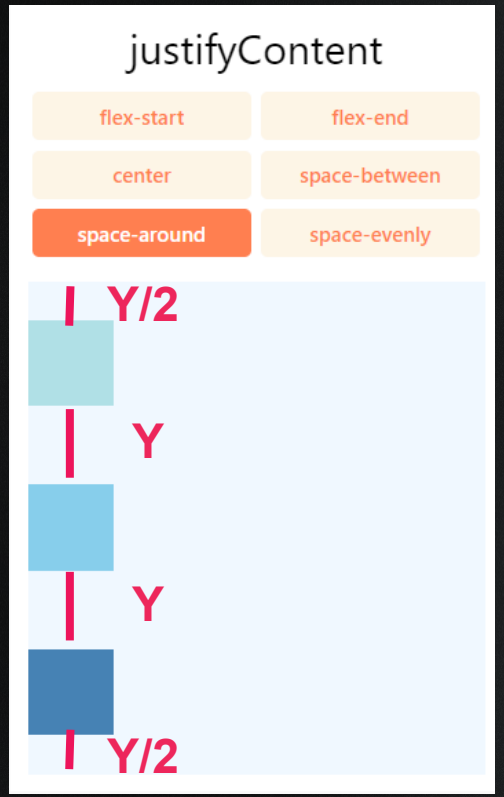

justifyContent
flex-start
flex-end
center
space-between
space-around
space-evenly

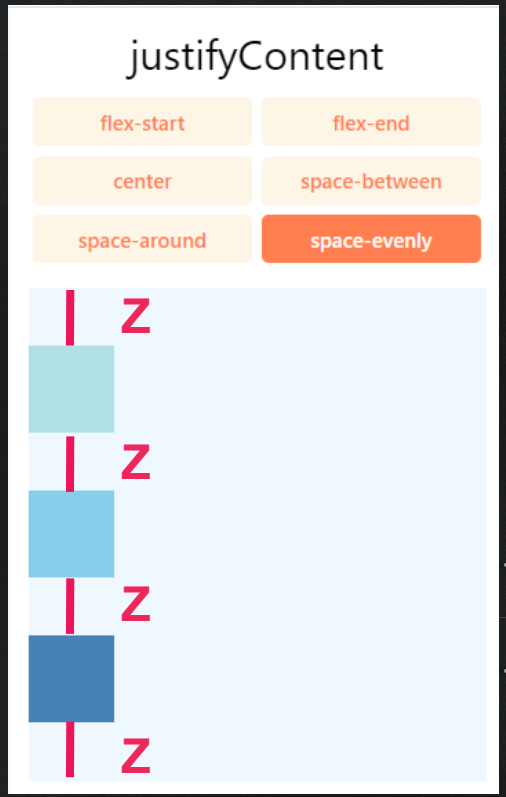
Flexbox


justifyContent
flex-start
flex-end
center
space-between
space-around
space-evenly


justifyContent
flex-start
flex-end
center
space-between
space-around
space-evenly
X
X

Flexbox


justifyContent
flex-start
flex-end
center
space-between
space-around
space-evenly
Y/2
Y
Y
Y/2


justifyContent
flex-start
flex-end
center
space-between
space-around
space-evenly
Z
Z
Z
Z

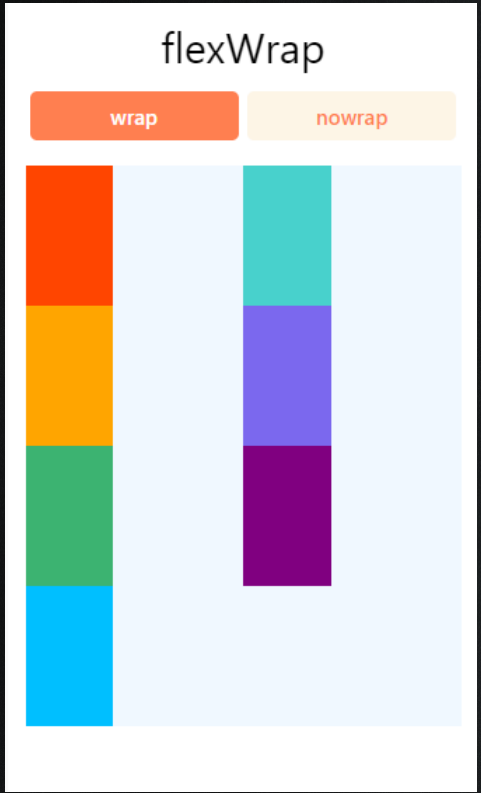
Flexbox

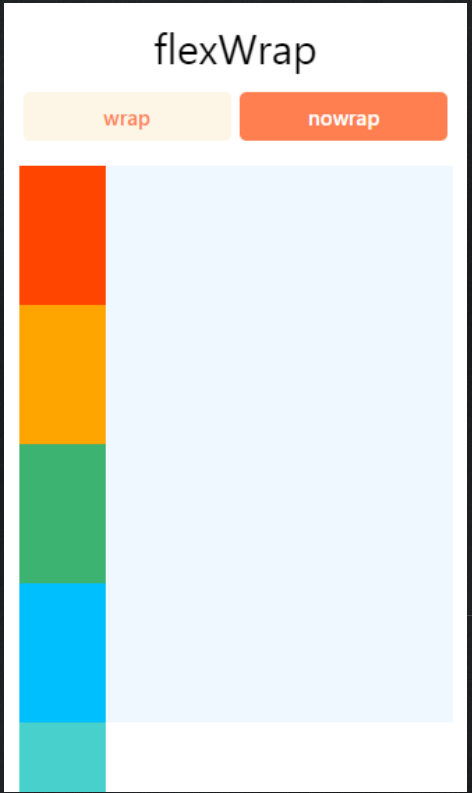
flexWrap

Determina se, ao ter mais itens que o tamanho do container, os itens irão “vazar” ou se irão formar uma nova linha ou coluna

- wrap
- **nowrap (default)**

Flexbox





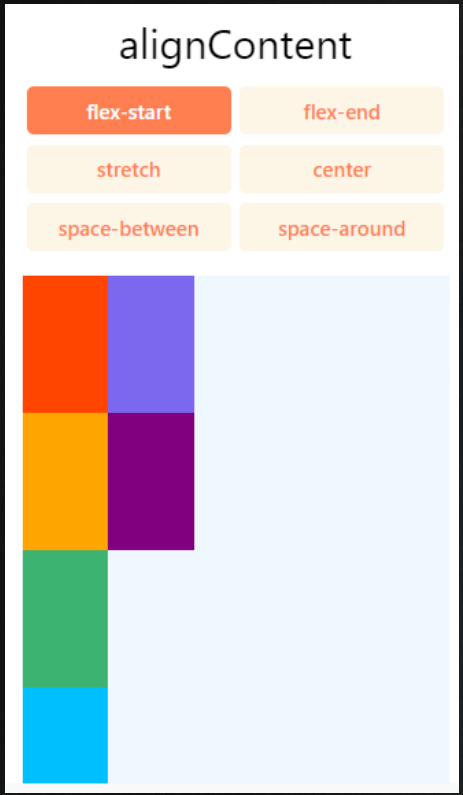
Flexbox

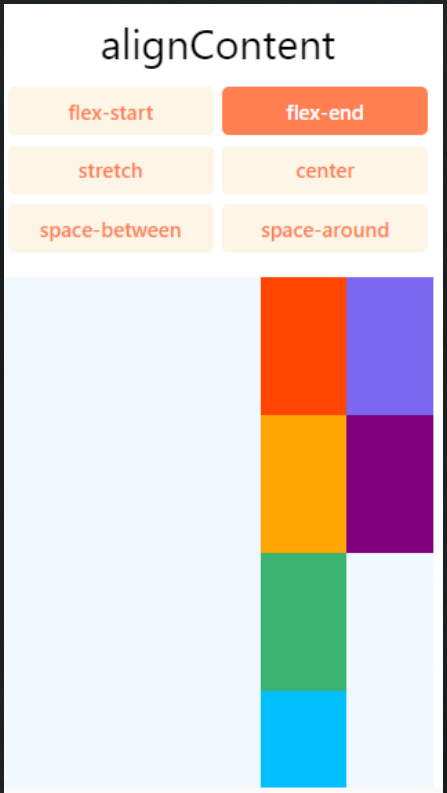
alignContent

Determina a distribuição das linhas/colunas quando usamos flexWrap: wrap

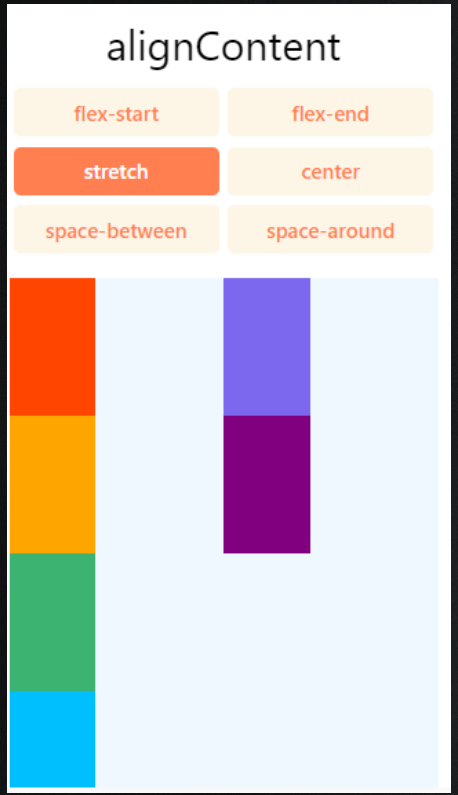
- **flex-start (default)**
- flex-end
- stretch
- center
- space-between
- space-around

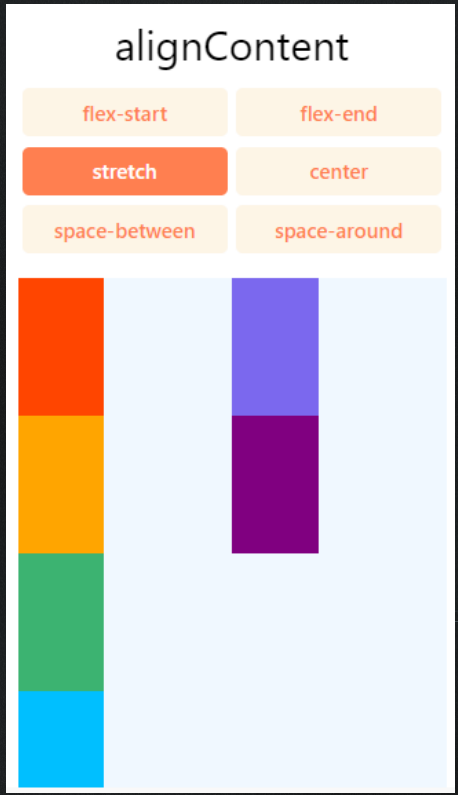
Flexbox


alignContent
flex-start
flex-end
stretch
center
space-between
space-around

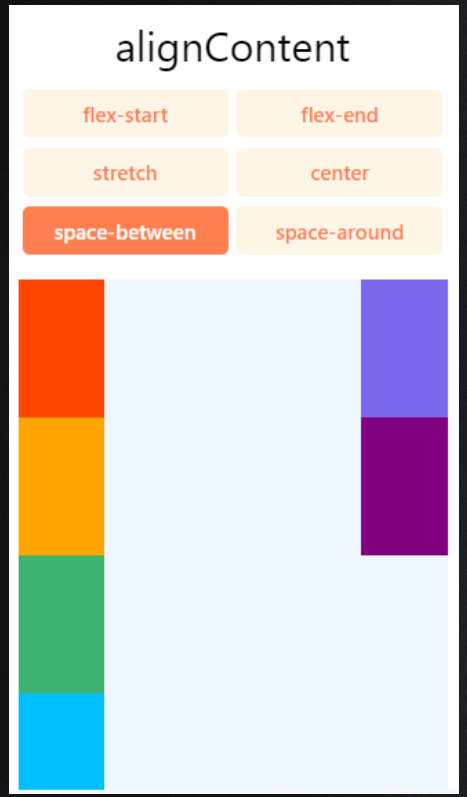

alignContent
flex-start
flex-end
stretch
center
space-between
space-around

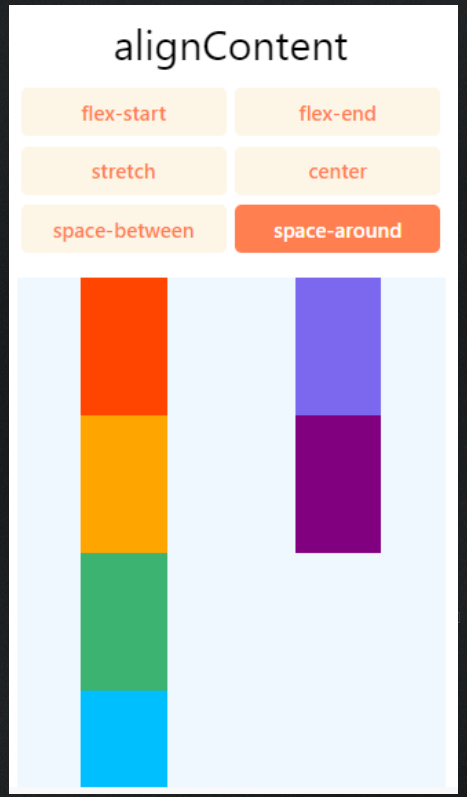
Flexbox


alignContent
flex-start
flex-end
stretch
center
space-between
space-around


alignContent
flex-start
flex-end
stretch
center
space-between
space-around

Flexbox


alignContent
flex-start
flex-end
stretch
center
space-between
space-around


alignContent
flex-start
flex-end
stretch
center
space-between
space-around

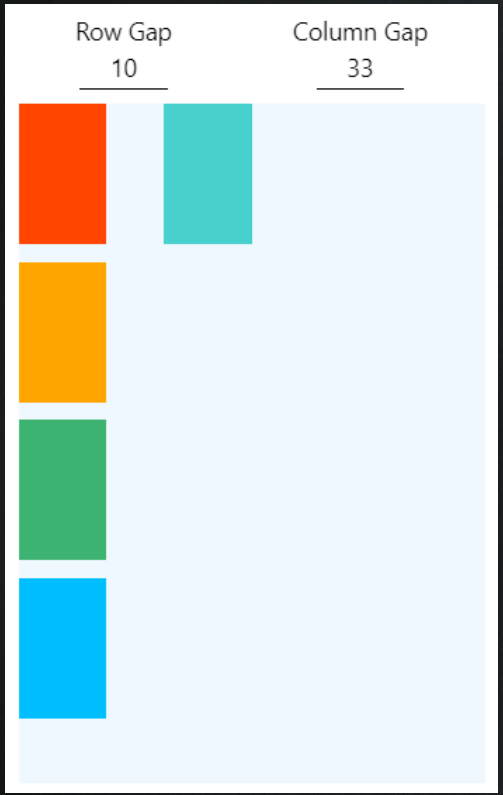
Flexbox

gap, rowGap, columnGap

Utilizado para determinar um valor fixo de distância entre os itens em um ou ambos os eixos

- **0 como default**

Flexbox



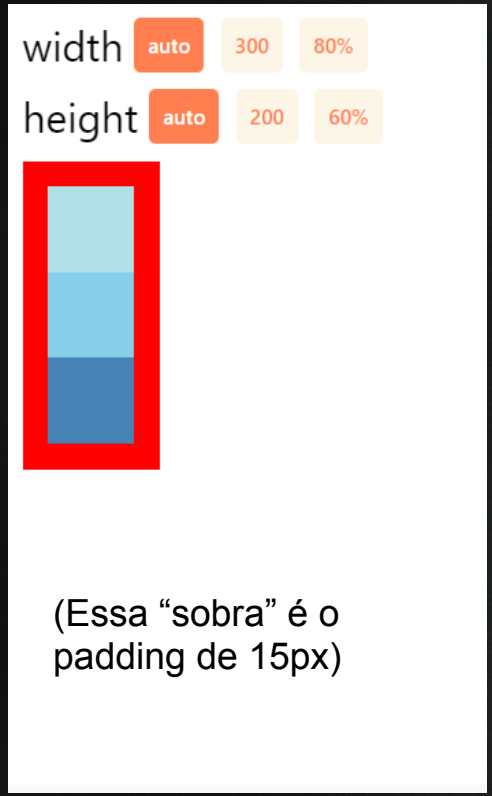
Flexbox

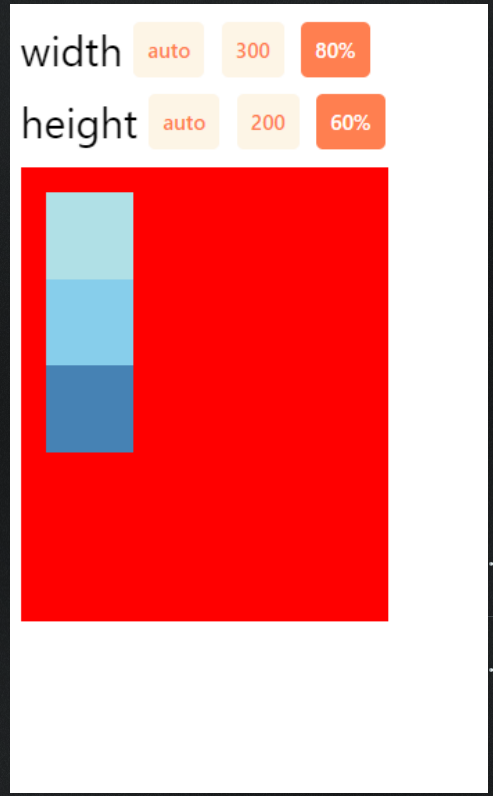
width e height

Determina altura e largura do seu item/container
O valor do container NÃO altera o valor de seus itens

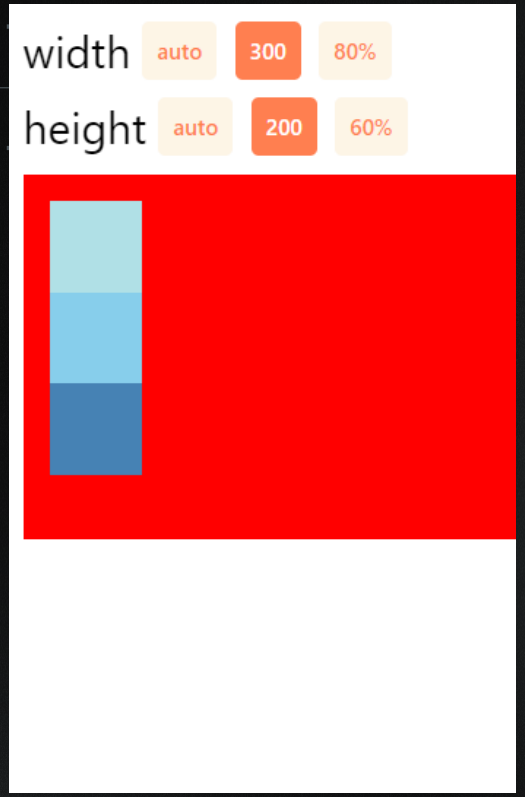
- **auto (default)**
- pixels
- porcentagem

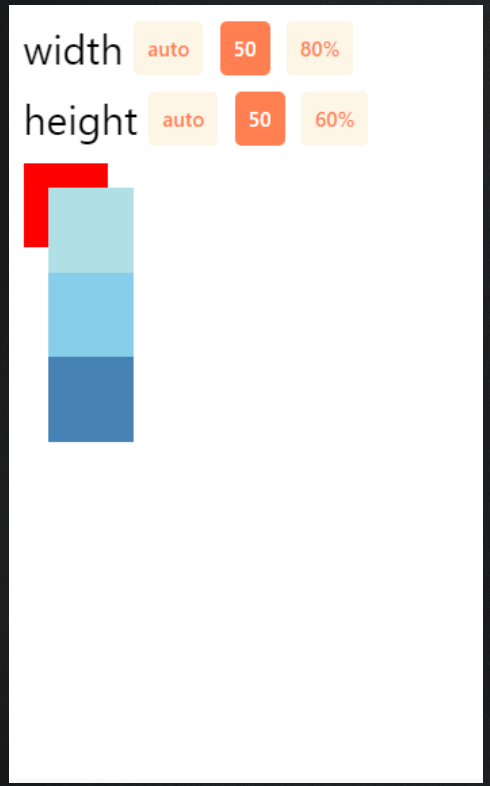
Flexbox

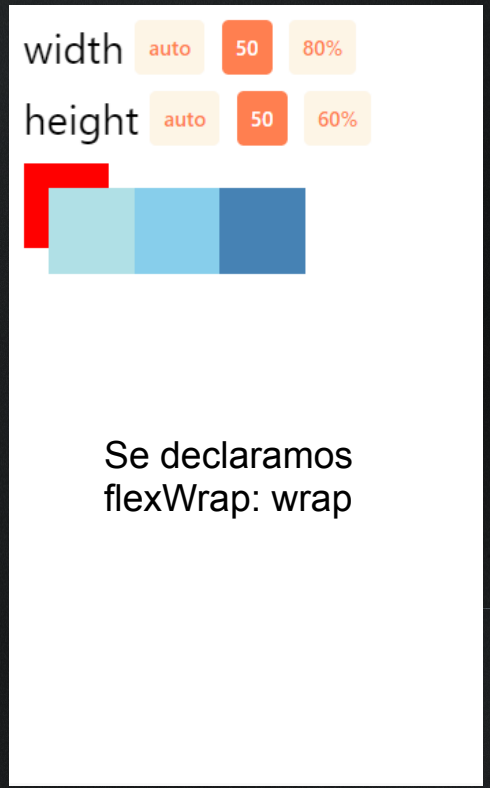

width
auto
300
80%
height
auto
200
60%
(Essa “sobra” é o padding de 15px)


width
auto
300
80%
height
auto
200
60%

Flexbox







Se declaramos flexWrap: wrap

Container

Flexbox

positioning

Determina a posição do item

- **relative (default)**
- absolute

- right/left/top/bottom **(default 0)**

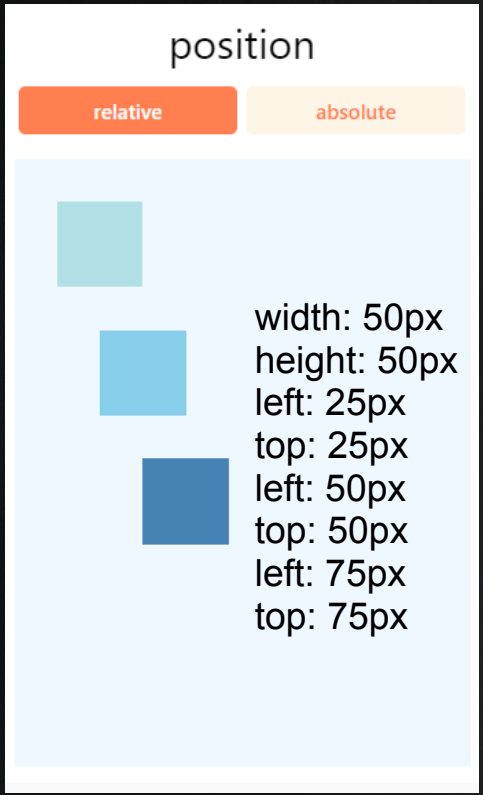
Flexbox

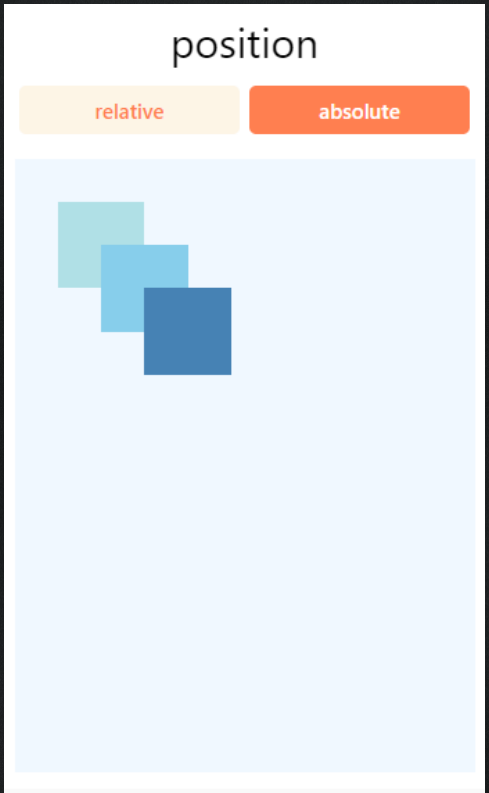
positioning

Para determinarmos a posição de um item, utilizamos top **ou** bottom, e left **ou** right.

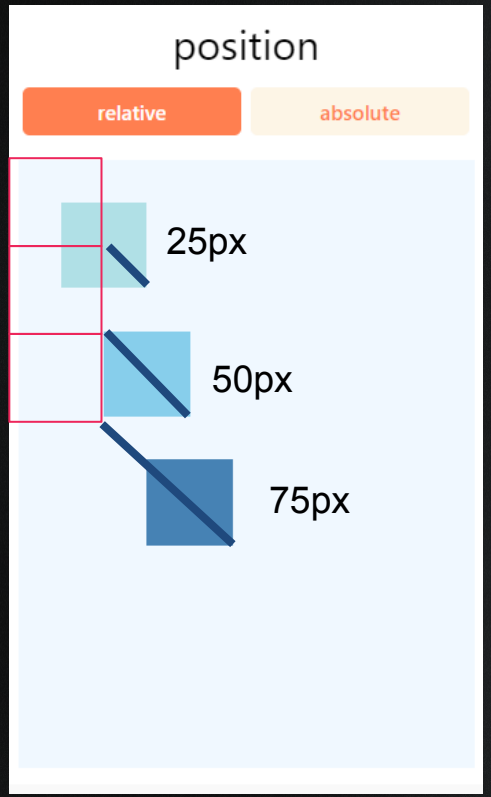
Não utilizamos posições que atuem no mesmo eixo juntos

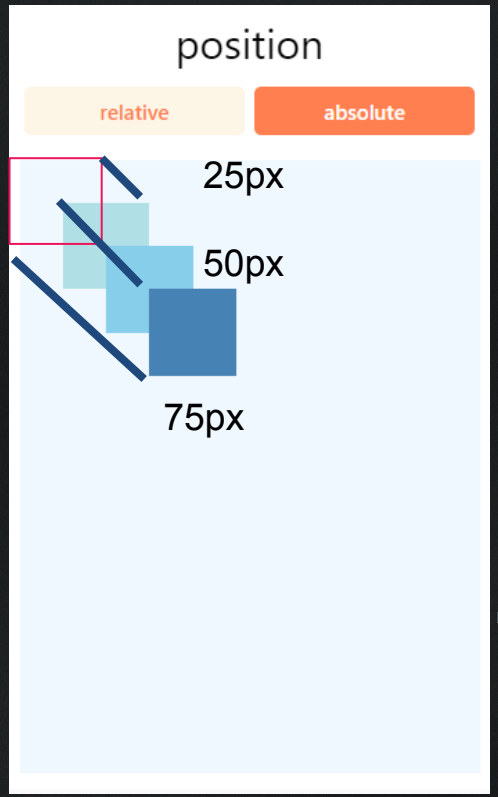
Flexbox


position
relative
absolute
width: 50px
height: 50px
left: 25px
top: 25px
left: 50px
top: 50px
left: 75px
top: 75px


position
relative
absolute

Flexbox


position
relative
absolute
25px
50px
75px


position
relative
absolute
25px
50px
75px

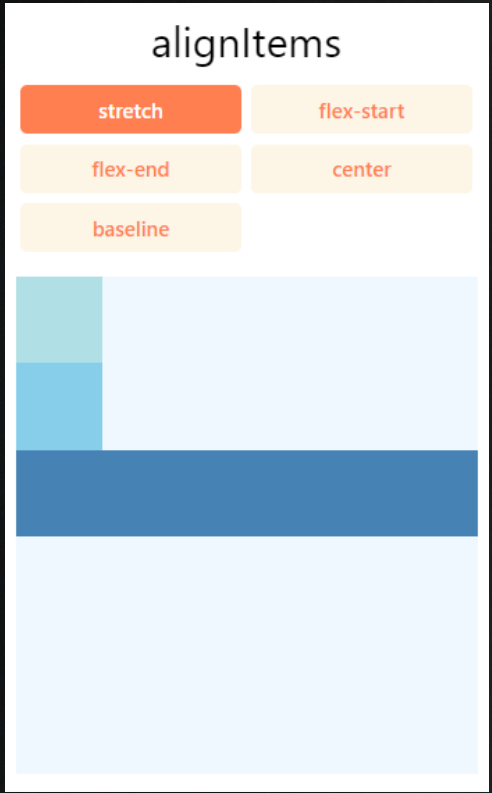
Flexbox

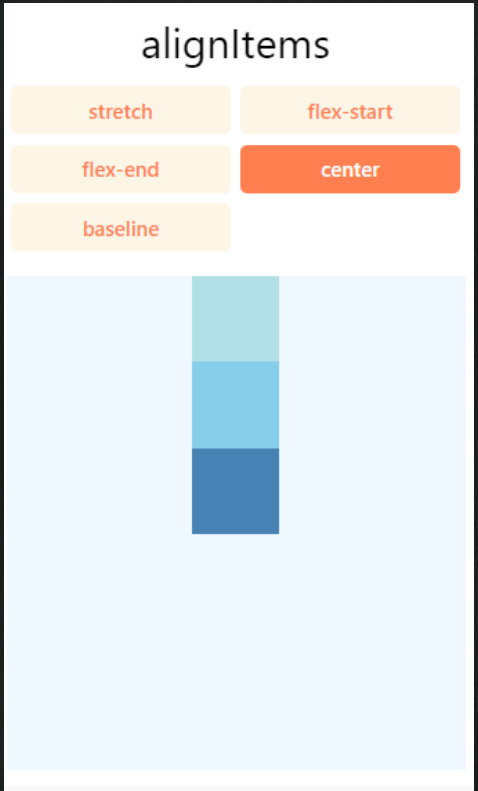
alignItems

É utilizado para alinhar os itens dentro do container. Semelhante ao justifyContent, porém o alignItems atua no **eixo transversal**, enquanto o justifyContent atua no **eixo principal**

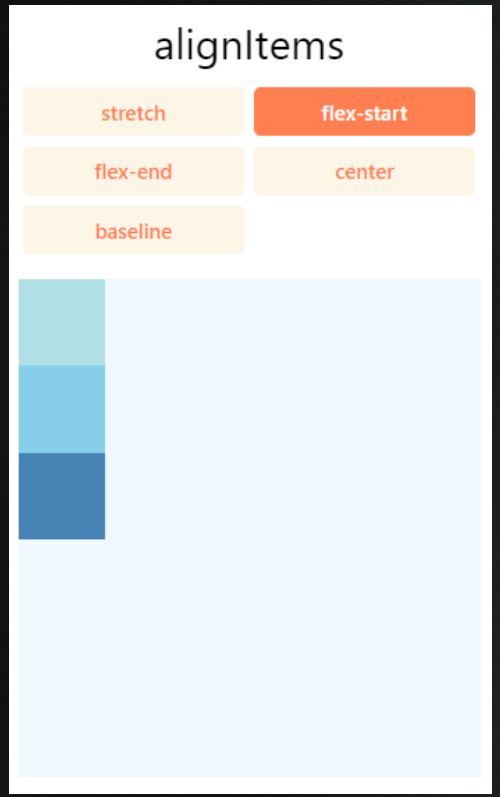
- **stretch (default)**
- flex-start
- flex-end
- center
- baseline

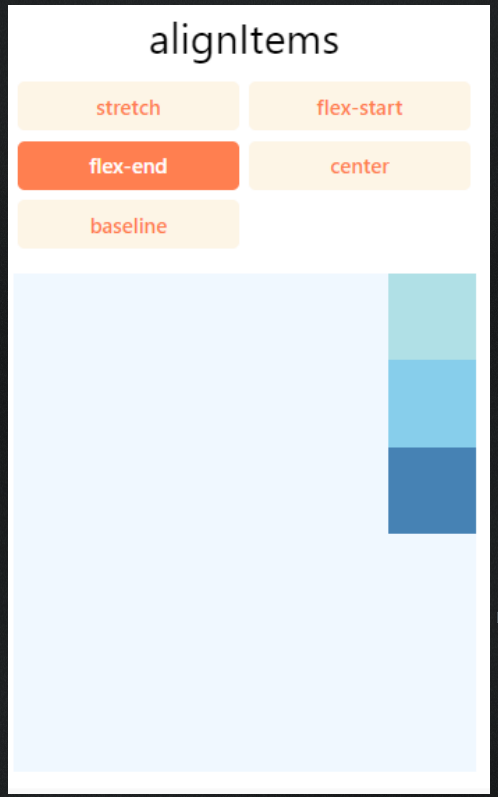
Flexbox


alignItems
stretch
flex-start
flex-end
center
baseline


alignItems
stretch
flex-start
flex-end
center
baseline

Flexbox


alignItems
stretch
flex-start
flex-end
center
baseline


alignItems
stretch
flex-start
flex-end
center
baseline

Flexbox

alignItems

O stretch só funciona quando não há um tamanho definido no eixo transversal. No caso anterior, 2 itens tinham width: 50px, apenas o item sem width que foi modificado

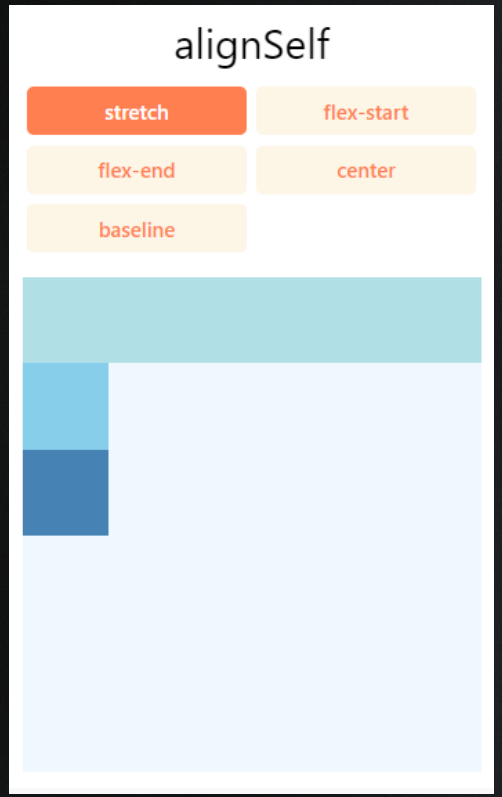
Flexbox

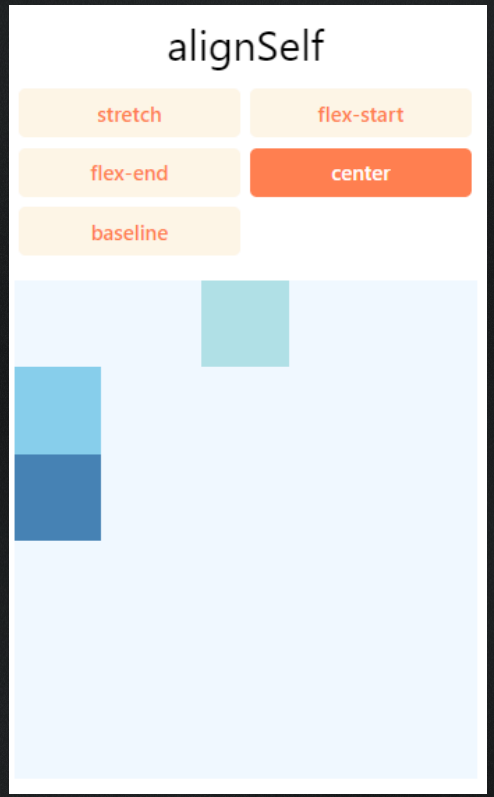
alignSelf

É semelhante ao alignItems, porém é aplicado no **item** e não no container. Se declarado, sobrescreve o alignItems

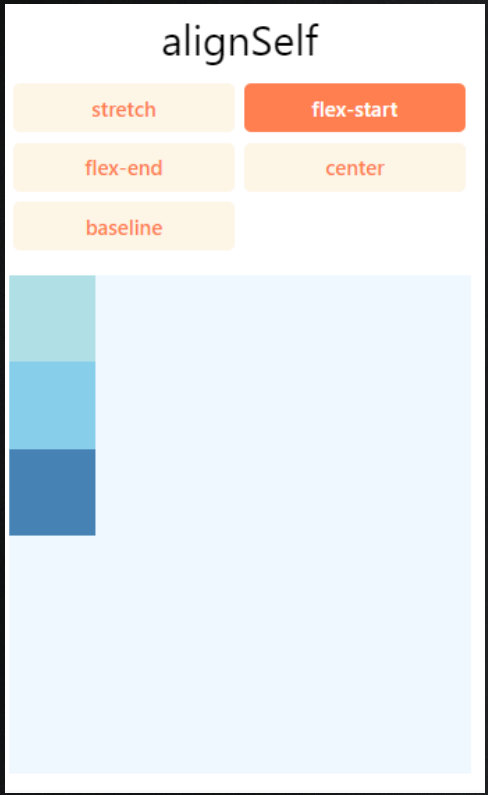
- **stretch (default)**
- flex-start
- flex-end
- center
- baseline

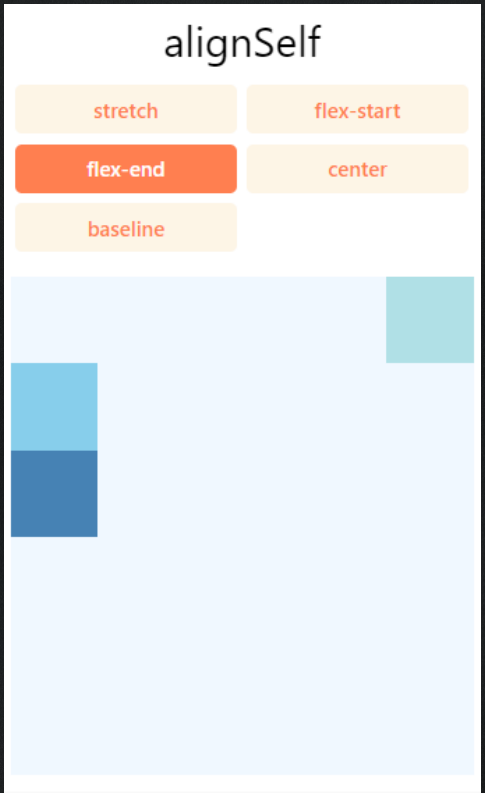
Flexbox


alignSelf
stretch
flex-start
flex-end
center
baseline


alignSelf
stretch
flex-start
flex-end
center
baseline

Flexbox


alignSelf
stretch
flex-start
flex-end
center
baseline


alignSelf
stretch
flex-start
flex-end
center
baseline

Flexbox

flexBasis, flexShrink, flexGrow

São parâmetros que definem como os itens irão se comportar para dividir o espaço do container entre si

- flexBasis: Define o valor “default” do item no eixo principal. Similar a height ou width
- flexShrink: Define o quanto um item irá diminuir para pode ocupar o espaço
- flexGrow: Define o quanto um item irá aumentar para ocupar o espaço disponível

Flexbox

flexBasis, flexShrink, flexGrow

FlexShrink é utilizado quando o espaço do container não é suficiente para todos os itens. Os itens são redimensionados para um tamanho menor, de acordo com o peso dado a cada um deles.

Flexbox

flexBasis, flexShrink, flexGrow

FlexGrow é utilizado quando o espaço do container é maior que o necessário para todos os itens. Os itens são redimensionados para um tamanho maior, de acordo com o peso dado a cada um deles.

Flexbox

flexBasis, flexShrink, flexGrow

flexBasis: O flexBasis default é o tamanho do item no eixo principal.
flexShrink: Valores positivos (float), com o default sendo 0
flexGrow: Valores positivos (float), com o default sendo 1

Flexbox

flex

flex é uma abreviação dos 3 parâmetros anteriores. Quando declaramos **flex**, estamos declarando flex-grow, flex-shrink, flex-basis nessa ordem

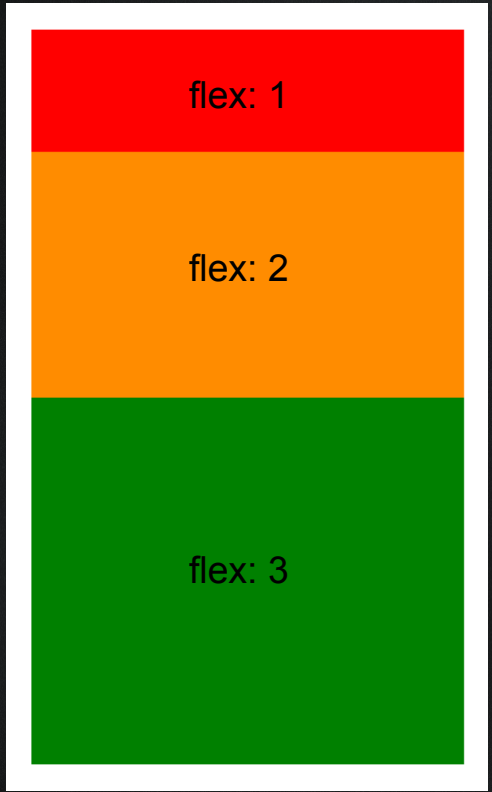
exemplos:
flex: 1 1 auto

Flexbox

flex (apenas 1 valor)

Quando utilizamos flex com apenas 1 valor, é o mesmo que declararmos **flex: 1 0 auto**
É muito comum utilizarmos apenas **flex: x**, para dividirmos o espaço entre os itens, já que é bem mais incomum uma necessidade de utilizar flexBaseline ou flexShrink

Flexbox



View

```
import React from 'react';
import {View, Text} from 'react-native';

const ViewBoxesWithColorAndText = () => {
  return (
    <View
      style={{
        flexDirection: 'row',
        height: 100,
        padding: 20,
      }}>
      <View style={{backgroundColor: 'blue', flex: 0.3}} />
      <View style={{backgroundColor: 'red', flex: 0.5}} />
      <Text>Hello World!</Text>
    </View>
  );
};

export default ViewBoxesWithColorAndText;
```

Style e Styles Sheet

Style e Stylesheet

O que é ao style?

É uma prop recebida pelos componentes para aplicar style CSS nos componentes react

- backgroundColor
- color
- width/height
- atributos flexbox
- Entre outros (alguns específicos para certos componentes)

Style e Stylesheet

Text

- fontFamily
- fontSize
- fontStyle
- fontWeight
- textAlign
- textDecorationLine

Style e Stylesheet

Image

- backgroundColor
- borderColor
- borderRadius
- opacity
- overflow
- resizeMode

Style e Stylesheet

Stylesheet

É um objeto especial, que encapsula e valida o estilo a ser aplicado no componente. Temos 2 principais métodos, o create e o compose.

- Create: cria um stylesheet a partir de um objeto
- Compose: cria um stylesheet a partir de 2 stylesheets, ou partes deles

Style e Stylesheet

Stylesheet

É um objeto especial, que encapsula e valida o estilo a ser aplicado no componente. Temos 2 principais métodos, o create e o compose.

- Create: cria um stylesheet a partir de um objeto
- Compose: cria um stylesheet a partir de 2 stylesheets, ou partes deles. Se houver repetição, o 1o stylesheet é sobrescrito pelo 2o

Style e Stylesheet

```
const page = StyleSheet.create({
  container: {
    flex: 1,
    padding: 24,
    backgroundColor: '#fff',
  },
  text: {
    fontSize: 30,
    color: '#000',
  },
});

const lists = StyleSheet.create({
  listContainer: {
    flex: 1,
    backgroundColor: '#61dafb',
  },
  listItem: {
    fontStyle: 'italic',
    fontWeight: 'bold',
  },
});

const container = StyleSheet.compose(page.container, lists.listContainer);
const text = StyleSheet.compose(page.text, lists.listItem);
```

Style Text

Style page.text

Style lists.listItem

Links

https://developer.mozilla.org/pt-BR/docs/Web/CSS/CSS_Flexible_Box_Layout/Basic_Concepts_of_Flexbox

https://reactnative.dev/docs/flexbox

https://css-tricks.com/snippets/css/a-guide-to-flexbox/

https://reactnative.dev/docs/stylesheet

Dúvidas, anseios, desabafos?

FIAP